

Module 1: Python Setup & Fundamentals

This module covers core programming fundamentals used in any software system, including AI applications. It builds the base for representing data and writing logic required for program execution and decision-making.

Topics

- Python Installation
- IDE Setup
- Virtual Environments
- Variables & Data Types
- Strings
- Lists
- Tuples
- Sets
- Dictionaries
- Conditional Statements
- Loops

Module 2: Implement Structured Python Code

This module covers topics on how to develop structured codebase in Python. These fundamentals are essential for building reliable and maintainable applications and form the basis for modular tools, workflow execution, and robust system behavior.

Topics

- Functions
- Import and Modules
- Exception Handling
- Object Oriented Programming

Module 3: Working with Files & Data

This module covers core Python fundamentals for handling external data sources such as files and datasets. These skills are essential for data ingestion, processing, and integration across applications and form the foundation for later use in RAG-based systems and Agentic AI workflows.

Topics

- File Operations
- CSV Files
- JSON Files
- PDF Files
- Directory & Path Handling

Module 4: APIs & Project Organization

This module covers working with external services using APIs and organizing Python projects in a structured way. It forms a foundational part of Python development required before working on LLM integrations and Agentic AI systems.

Topics

- API Fundamentals
- API Requests
- API Response Handling
- Modules
- Packages
- Environment Variables

Module 5: Modern Python for AI Development

This module focuses on advanced Python capabilities essential for building scalable Agentic AI systems, especially structured outputs and concurrent execution of AI workflows.

Topics

- CLI Script
- Type Hints
- Async Fundamentals
- Asyncio
- Concurrent Execution